Department of Numerical Analysis and Scientific Computing
Simula Research Laboratory
Oslo, Norway

Optimization in Oslo

A Seminar Series on Continuous Optimization

Date:
Wednesday November 22, 2023 at 14:00 (CET)

Speaker:
Prof. Carlos Rautenberg
George Mason University

Title:
Non-diffusive Variational Problems with Distributional and Weak Gradient Constraints

Abstract:
We consider non-diffusive variational problems with mixed boundary conditions and with distributional or weak gradient constraints. This class of problems arises naturally in the modeling of the growth of non-homogeneous sand piles and the problem is posed in spaces of functions of bounded variation or Sobolev spaces. In this setting, the upper bound in the constraint is either a function or a Borel measure which allows the pile growth to observe discontinuities. We address existence and uniqueness of the model under low regularity assumptions, and rigorously identify its Fenchel pre-dual problem. The latter in some cases is posed on a non-standard space of Borel measures with square integrable divergences. Further, we investigate the perturbation of solutions to the problem by means of perturbation of the upper bound of the gradient constraint and establish stability results. We conclude the talk by introducing a mixed finite-element method and several numerical tests.

Brief Bio:
Carlos N. Rautenberg is an associate professor in the Mathematical Sciences Department at George Mason University. Prior to joining George Mason, he was an ECMath (Einstein Center for Mathematics Berlin) Junior Research Group Leader at the Department of Mathematics of the Humboldt Universität

zu Berlin, and the former deputy head of Research Group 8 at Weierstrass Institute for Applied Analysis and Stochastics in Berlin, Germany. His research interests involve several areas of analysis and applied mathematics: optimal control, variational and quasi-variational inequalities, optimal filtering, and mathematical imaging.